



A Comparative Analysis of Large Language Model(LLM) and Human-Generated Text Using Natural Language Processing

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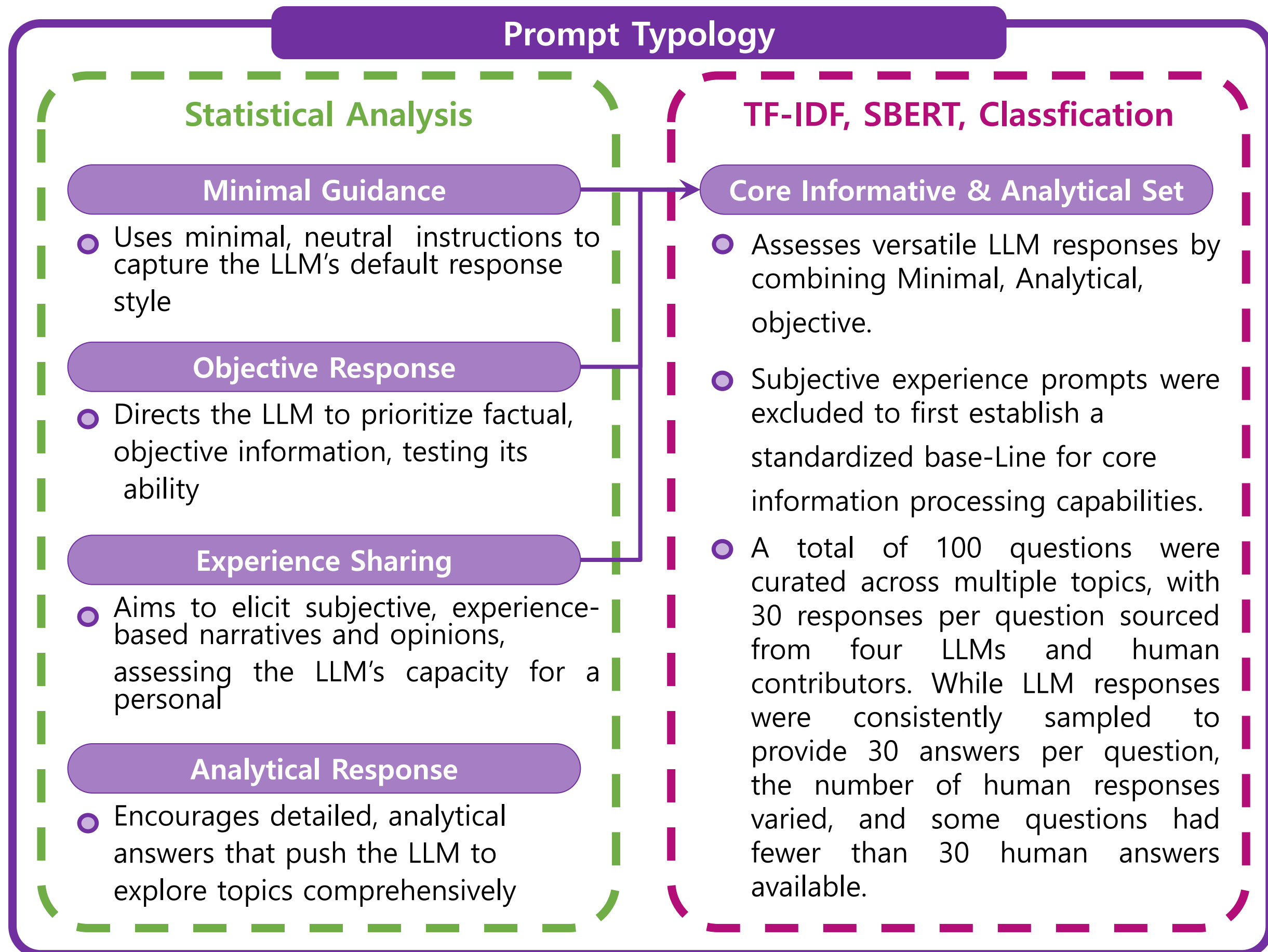
Introduction

Key question: Can we still tell apart human and machine-generated text?

- This study explores “Differences” between human responses and those from major LLMs
- Modern LLMs(e.g., GPT, Gemini) produce text nearly indistinguishable from human writing
- Focused on outputs from Claude, DeepSeek, Gemini, GPT and Human
- Aims to identify “syntactic and lexical Fingerprints” for each source
- Investigates whether yet classifier works to discriminate the human despite fluent coherent writing of LLM

Data & Methodology

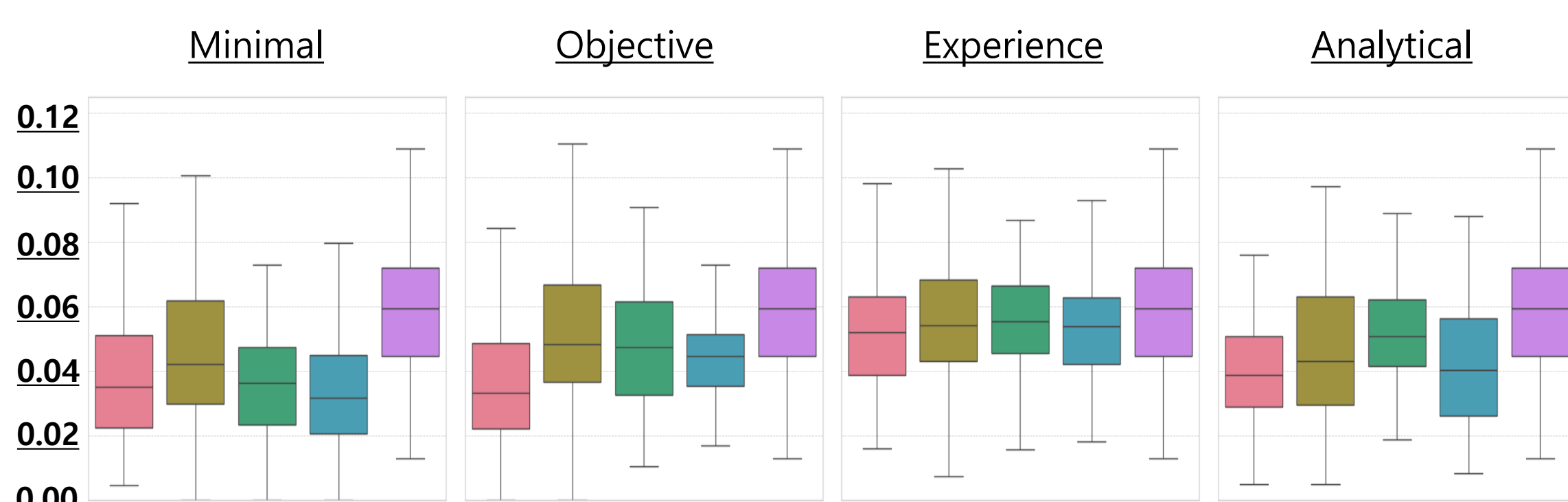
- First, for the purpose of analyzing and classifying responses as either LLM-generated or human-written, we collected real-world answers from human users on Quora and generated responses from four major LLMs: Claude, Gemini, DeepSeek and GPT
- Second, we used five distinct prompt modes for varied LLM response characteristics
- We explored various linguistic features (e.g., syntactic stats, POS tags, TF-IDF, SBERT, LDA)
- For classification, we trained a LightGBM model using input features based on LDA-based subtopic distributions.



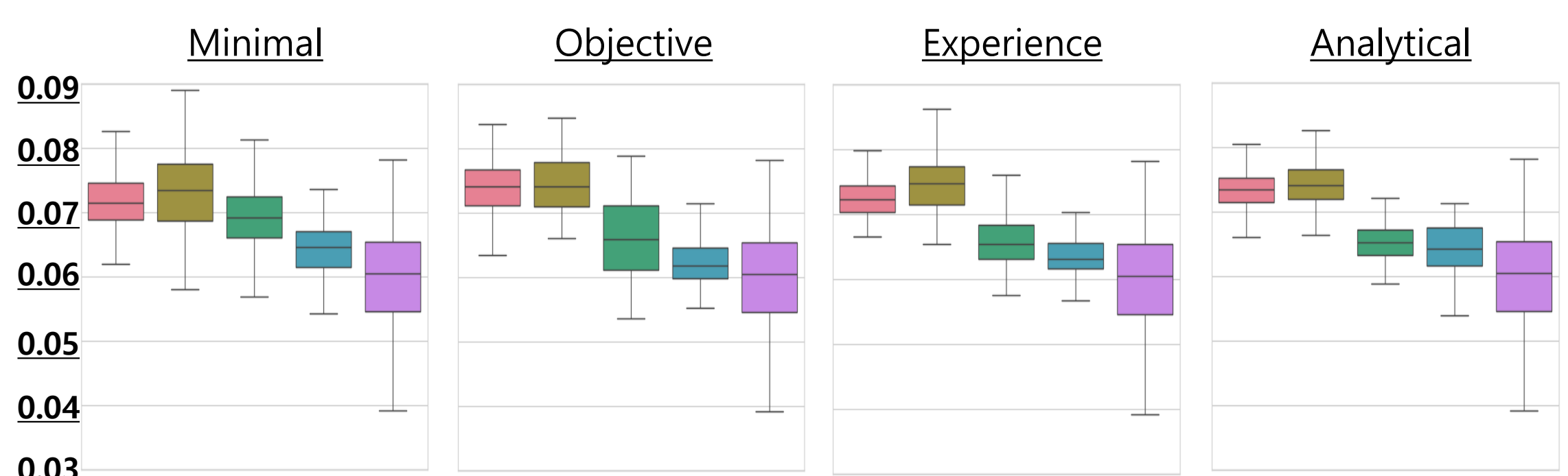
- Lastly, the input texts were cleaned by removing special characters, lowercased, tokenized using NLTK(Natural Language ToolKit), and filtered to exclude English stopwords for data preprocessing & tokenization
- Linguistic analysis(stylometry, POS) clearly distinguishes human text from LLMs. Humans show vast stylistic diversity. LLMs exhibit more consistent, model-specific signatures.

Syntactic Fingerprinting : How They Speak

Personal Pronoun Proportion



Lexical Density



Claude **DeepSeek** **Gemini** **GPT** **Human**

$$\text{Lexical Density} = \frac{\text{Number of Unique Words}}{\text{Total Number of Words(Tokens)}} \quad \text{PRP Proportion} = \frac{\text{Number of Personal Pronouns}}{\text{Total Number of Words}}$$

Topical & Lexical Signatures : What They Speak about

Term Frequency - Inverse Document Frequency

- LLMs generate responses using a common, general-purpose lexicon. In contrast, human lexical choices indicate responses more grounded in specific contexts, personal perspectives, temporal awareness

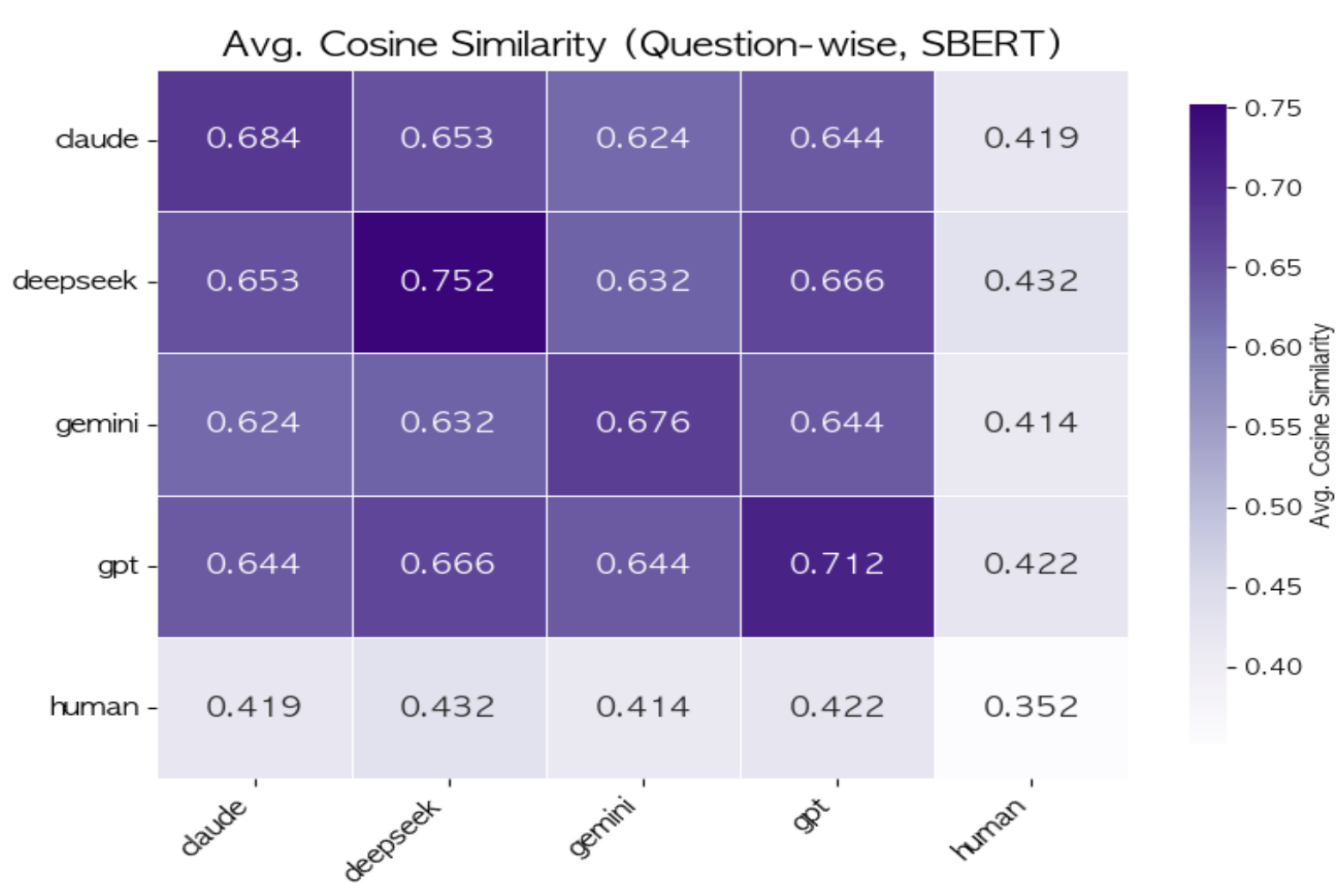
Top-10	Human		GPT		Gemini		DeepSeek		Claude	
	Word	Score	Word	Score	Word	Score	Word	Score	Word	Score
1	One	0.2570	Like	0.3072	Like	0.3796	Like	0.3088	Like	0.2530
2	Time	0.2424	One	0.2373	Think	0.2711	One	0.2925	People	0.2247
3	Would	0.2235	Time	0.1840	Thing	0.2285	People	0.2696	I've	0.2189
4	People	0.1879	People	0.1837	One	0.1966	Even	0.2426	Think	0.2042
5	Like	0.1807	Think	0.1768	People	0.1805	Way	0.1815	Something	0.1804
6	Get	0.1582	Might	0.1735	Even	0.1754	Thing	0.1621	One	0.1750
7	Day	0.1550	Thing	0.1595	Okay	0.1751	Time	0.1506	Actually	0.1681
8	Year	0.1521	I've	0.1563	Really	0.1610	Make	0.1402	Thing	0.1529
9	Life	0.1457	Even	0.1517	Know	0.1607	Something	0.1389	Time	0.1517
10	dont	0.1381	Life	0.1473	something	0.1361	dont	0.1331	someone	0.1344

$$TF - IDF = TF * IDF$$

- TF(Term Frequency) : How often a word appears in a document
- IDF(Inverse Document Frequency) : How rare a word is across all documents

Sentence BERT

- SBERT (sentence embedding) analysis shows LLMs are highly similar and consistent in their semantic meaning.



- Human responses are distinctly different from LLMs and far more internally diverse

Classification of LLM and Human Text

Hierarchical LDA-Based Subtopic Modeling

- Topic Modeling (LDA)
 - Construct Dictionary and Bag-of-Words from tokenized data
 - The number of topics was set to five (four LLMs and one human)
 - Represent each document as a distribution over main topics
- Subtopic Modeling
 - Select top 20 representative documents for each main topic
 - Identify optimal number of subtopics within each document group
 - Apply subtopic model to all documents to compute subtopic probability distributions

Doc_id	0-1	0-2	1-1	1-2	1-3	1-4	...	LLM_Name
1	0.7245	0.2755	0.1853	0.2493	0.2679	0.2975	...	GPT-4
2	0.9074	0.0926	0.2296	0.2947	0.1642	0.3115	...	Claude
3	0.8692	0.1308	0.2779	0.2186	0.2379	0.2656	...	Claude
4	0.8739	0.1261	0.1452	0.2068	0.3081	0.3399	...	Deepseek
...								
n-1	0.8486	0.1514	0.2454	0.2154	0.3099	0.2292	...	Human
n	0.8790	0.1210	0.1651	0.2923	0.2491	0.2935	...	Human

* This matrix is used as the input feature set for classification

LightGBM Based Classification

- Input X : Subtopic probability vector per document
- Target Y : LLM name associated with each document
- Classifier : LightGBM multi-class model (Splitting Train 60% and Test 40%)

	precision	recall	f1-score	support
claude	0.614907	0.66	0.636656	1200
deepseek	0.679865	0.6725	0.676163	1200
gemini	0.75136	0.805833	0.777644	1200
gpt	0.805461	0.786667	0.795953	1200
human	0.86	0.758377	0.805998	1134
accuracy	0.736434			
Macro avg	0.742319	0.736675	0.738483	5934
Weighted avg	0.74101	0.736434	0.737732	5934

LightGBM Confusion Matrix

True Label \ Predicted Label	claude	deepseek	gemini	gpt	human
claude	792	189	103	56	60
deepseek	207	807	68	60	58
gemini	97	56	967	70	10
gpt	83	61	100	944	12
human	109	74	49	42	860

- The LightGBM model had an overall accuracy of 73.6%, classifying human responses most accurately, while Claude and DeepSeek were often confused. In particular, Gemini and GPT had high f1-scores, showing that they reflect the differences in responses between LLMs.

Conclusion & Future Work

- This study demonstrates that distinctive linguistic fingerprints exist, enabling reliable classification between human and LLM-generated text, as well as among different LLMs. These differences stem from core features like semantic diversity and narrative style.
- The classifier developed here can help combat Model Collapse, the degradation of model trained on AI generated content. By assigning an “AI-generated probability score” to training data, it can assist in filtering such content, preserving data diversity and ensuring more sustainable LLM development.

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